Assessment Series. -Day 1(B)

Contest Problem Report

Contest Code	HUAGF
Total Problems	5
Coding Problems	5

Problem Name: Query over Matrix

Problem Statement:

There is a **N*N** square matrix. You have been given **N** specifications, one per row of the matrix. The **i^{th}** specifications is represented by a pair of integers **L[i]** and **R[i]**, denoting that all element belonging to columns between **L[i]** and **R[i]** (both inclusive) in row **i** are 1, while the rest of the elements of the **i^{th}** row are 0.

Now you will be given an integer **Q**, followed by **Q** operations: Each query is described by two integers **x** and **y**, which asks what will be the parity of the sum of the matrix if the **x^{th}** row and the **y^{th}** column are removed from the matrix.

Input

- First line of the input contains an integer **N**, followed by **N** lines: each containing a pair of integers. **i^{th}** pair of integers denotes the value of **L[i]** and **R[i]**.
- The next line contains the value of **Q**, and the following **Q** lines contain the queries.

Output

Q lines, each containing a single character "E" or "O" (no quotes needed),indicating whether the sum is even or odd.

Constraints

Subtask #1: 20 points

- **$1 \le N,Q \le 1000$, $1 \le L[i] \le R[i] \le N$,$1 \le x,y \le N$**

Subtask #2: 80 points

- **$1 \le N,Q \le 10^5$, $1 \le L[i] \le R[i] \le N$, $1 \le x,y \le N$**

Sample Input 1:

3 1 3 2 2 1 3 2 2 2 3 2

Sample Output 1:

E E

Explanation:

The given array is:
1 1 1
0 1 0
1 1 1
It's easy to see that the sum is even in both the queries.

Problem Name: Many Chefs

Problem Statement:

Chef Ciel wants to put a fancy neon signboard over the entrance of her restaurant. She has not enough money to buy the new one so she bought some old neon signboard through the internet. Ciel was quite disappointed when she received her order - some of its letters were broken. But she realized that this is even better - she could replace each broken letter by any letter she wants. So she decided to do such a replacement that the resulting signboard will contain the word **"CHEF"** as many times as possible.

We can model the signboard as a string **S** having capital letters from **'A'** to **'Z'**, inclusive, and question marks **'?'**. Letters in the string indicate the intact letters at the signboard, while question marks indicate broken letters. So Ciel will replace each question mark with some capital letter and her goal is to get the string that contains as many substrings equal to **"CHEF"** as possible. If there exist several such strings, she will choose the lexicographically smallest one.

Note 1. The string $\mathbf{S = S_1...S_N}$ has the substring **"CHEF"** if for some **i** we have $\mathbf{S_iS_{i+1}S_{i+2}S_{i+3}}$ **= "CHEF"**. The number of times **"CHEF"** is the substring of **S** is the number of those **i** for which $\mathbf{S_iS_{i+1}S_{i+2}S_{i+3}}$ **= "CHEF"**.

Note 2. The string $\mathbf{A = A_1...A_N}$ is called lexicographically smaller than the string $\mathbf{B = B_1...B_N}$ if there exists **K** from **1** to **N**, inclusive, such that $\mathbf{A_i = B_i}$ for **i = 1, ..., K-1**, and $\mathbf{A_K < B_K}$. In particular, **A** is lexicographically smaller than **B** if $\mathbf{A_1 < B_1}$. We compare capital letters by their positions in the English alphabet. So **'A'** is the smallest letter, **'B'** is the second smallest letter, ..., **'Z'** is the largest letter.

Input

The first line of the input contains an integer **T** denoting the number of test cases. The description of **T** test cases follows. The only line of each test case contains a string **S**.

Output

For each test case, output a single line containing the content of the signboard Chef Ciel will come up with. That is you should output the lexicographically smallest string that could be obtained from the input string by replacing all its question marks by some capital letters and having as many substrings equal to **"CHEF"** as possible.

Constraints

- **1** ≤ **T** ≤ **2013**
- **1** ≤ length of **S** ≤ **2013**
- Each character in **S** is either a capital letter from **'A'** to **'Z'**, inclusive, or the question mark **'?'**.

Sample Input 1: 5 ????CIELIS???E? ????CIELISOUR???F T?KEITE?SY ???????? ???C??? ### Sample Output 1: CHEFCIELISACHEF CHEFCIELISOURCHEF TAKEITEASY CHEFCHEF AAACHEF ### Explanation:

Example Case 1. Here the resulting string can have at most 2 substrings equal to **"CHEF"**. For example, some possible such strings are:

- **CHEF**CIELISA**CHEF**
- **CHEF**CIELISQ**CHEF**

- **CHEF**CIELISZ**CHEF**

However, lexicographically smallest one is the first one.

Example Case 3. Here the resulting string cannot have **"CHEF"** as its substring. Therefore, you must simply output the lexicographically smallest string that can be obtained from the given one by replacing question marks with capital letters.

Problem Name: Palindromes Not Allowed

Problem Statement:

A string S is called *good* if none of the substrings of S of length ≥ 2 are palindromic.

For e.g. $S = \texttthello$ is not good but $S = \textttworld$ is good.

You are given a string A of length N (containing lowercase Latin letters only). Rearrange A to form a good string or determine it is not possible to do so.

Recall that a string is called palindromic if it reads the same backwards and forwards. For example, $\texttttnoon$ and $\textttlevel$ are palindromic strings.

Input Format

- The first line of input contains a single integer T — the number of test cases. The description of the test cases follows.
- The first line of each test case contains one integer N — the length of the string A.
- The second line of each test case contains the string A of length N.

Output Format

For each test case, if it is possible to rearrange A to convert it into a good string, print YES in the first line and print the rearranged string in the second line. Otherwise, print NO.

You may print each character of YES and NO in uppercase or lowercase (for e.g. yes, yEs, Yes will be considered identical).

If there are multiple rearrangements of the string which are good, **print any of them**.

Constraints

- $1 \leq T \leq 10^5$
- $1 \leq N \leq 10^5$
- The string A contains lowercase Latin letters only
- The sum of N over all test cases does not exceed $2 \cdot 10^5$

Subtasks

Sample Input 1:

4 5 world 5 hello 7 aaabbcc 6 pppppr

Sample Output 1:

YES world YES lehol YES abcabca NO

Explanation:

Test case 1: $\textttworld$ is already a good string.

Test case *2*: *texttthello* can be rearranged to form *textttlehol* which is good.

Test case *3*: *textttaaabbcc* can be rearranged to form *textttabcabca* which is good.

Test case *4*: It can be proven that none of the rearrangements of *textttpppppr* is good.

Problem Name: Chef and Close Friends

Problem Statement:

Chef lives at position x on the x - axis.

Chef has $2y$ friends, each living at every integer point in the range: $[x - y, x + y]$ except the position x itself.

Chef wants to visit his friends, but his mother has placed a strict rule: Chef is allowed to travel at most z units away from his home in either direction. This means Chef can only move within the interval $[x - z, x + z]$.

Your task is to determine how many of Chef's friends live within the range Chef is allowed to travel.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of a single line of input. The line contains three space-separated integers x, y, and z — denoting Chef's position on the x-axis, the friend range $[x - y, x + y]$, and the maximum distance Chef is allowed to travel.

Output Format

For each test case, output a single integer — the number of friends Chef can visit.

Constraints

- $1 \leq T \leq 3 \times 10^5$
- $-50 \leq x \leq 50$
- $0 \leq y, z \leq 50$

Sample Input 1:

4 0 2 1 10 3 5 -5 4 2 7 10 0

Sample Output 1:

2 6 4 0

Explanation:

- In the 1st test case, chef's friends live at position [-2, 2] except 0 and chef can visit positions [-1, 1]. So he can visit 2 friends.
- In the 4th test case, chef can only visit position 7. So he cannot travel to any of his friends.

Problem Name: Fake Certificate

Problem Statement:

You are given a binary string S of length N, representing the attendance sheet of Alice.
For the days when Alice went to school, $S_i = 1$, otherwise, $S_i = 0$.

Alice wanted to increase her attendance. She can perform the following operation **at most** once :

- Choose any substring of S where Alice was absent **everyday**. She can then give a medical certificate for this period and will be marked present for the whole period .

Note that a substring is a contiguous subsegment of a string. For example, `acab` is a substring of `abacaba`, but `aa` or `d` aren't substrings of this string. In other words, substring $S[l, r] = S_lS_{l+1}...S_r$.

You need to find the **maximum** number of days Alice will be marked present after performing the operation **at most** once .

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.
  - The first line of each test case contains an integer N — the length of the string .
  - The second line of each test case contains a binary string S.

Output Format

For each test case, output on a new line the maximum number of days Alice will be marked present after at most one operation.

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq N \leq 2 \cdot 10^5$
- S is a Binary String
- The sum of N over all test cases won't exceed $2 \cdot 10^5$.

Sample Input 1:

4 3 111 3 000 6 010010 6 001001

Sample Output 1:

3 3 4 4

Explanation:

Test case 1: Alice was present on all days so no operation is needed. Maximum number of days Alice will be marked present is 3.

Test case *2*: Alice was absent on all days so she can choose substring *S[1,3]* and convert all absent to present. Maximum number of days Alice will be marked present is *3*.

Test case *3*: Alice can choose from *S[1,1]*, *S[3,4]* and *S[6,6]*. It is optimal to choose *S[3,4]* and the resulting string would be *011110*. Maximum number of days Alice will be marked present is *4*.